MAT 314 HW08

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Due 05/08

Problem. 1.1. Determine the orbit of the polynomial below. If the polynomial is symmetric, write it in terms of the elementary symmetric functions.

(a) $u_1^2u_2 + u_2^2u_3 + u_3^2u_1$, $n = 3$

(b) $(u_1 + u_2)(u_2 + u_3)(u_3 + u_1)$, $n = 3$

(c) $(u_1 - u_2)(u_2 - u_3)(u_1 - u_3)$, $n = 3$

(d) $u_1^3u_2 + u_2^3u_3 + u_3^3u_1 - u_1u_2^3 - u_2u_3^3 - u_3u_1^3$, $n = 3$

(e) $u_1^3 + u_2^3 + \cdots + u_n^3$

Solution. (a) Since cycling $u_1 \to u_2 \to u_3 \to u_1$ preserves the polynomial, even permutations (of which there are 3) preserve the polynomial. Thus the size of the orbit is $6/3 = 2$, and the two polynomials in the orbit are

$$u_1^2u_2 + u_2^2u_3 + u_3^2u_1, \quad u_1u_2^2 + u_2u_3^2 + u_3u_1^2$$

with the latter obtained by the permutation (1 2).

(b) This is symmetric: expanding shows that it's equal to

$$\begin{aligned} u_1^2u_2 + u_2^2u_3 + u_3^2u_1 + u_1u_2^2 + u_2u_3^2 + u_3u_1^2 + 2u_1u_2u_3 &= (u_1 + u_2 + u_3)(u_1u_2 + u_2u_3 + u_3u_1) - u_1u_2u_3 \\ &= \boxed{s_1s_2 - s_3}. \end{aligned}$$

(c) Expanding gives

$$u_1^2u_2 + u_2^2u_3 + u_3^2u_1 - u_1u_2^2 + u_2u_3^2 + u_3u_1^2.$$

As with (a) there are two polynomials in the orbit, which are

$$\begin{gathered} u_1^2u_2 + u_2^2u_3 + u_3^2u_1 - u_1u_2^2 + u_2u_3^2 + u_3u_1^2 \\ u_2^2u_1 + u_3^2u_2 + u_1^2u_3 - u_2u_1^2 + u_3u_2^2 + u_1u_3^2 \end{gathered}$$

(d) Same as (a), we can conclude that there are only two polynomials in the orbit, and they are

$$\begin{gathered} u_1^3u_2 + u_2^3u_3 + u_3^3u_1 - u_1u_2^3 - u_2u_3^3 - u_3u_1^3 \\ u_2^3u_1 + u_3^3u_2 + u_1^3u_3 - u_2u_1^3 - u_3u_2^3 - u_1u_3^3 \end{gathered}$$

(e) This is symmetric. We first note that

$$u_1^2 + u_2^2 + \cdots + u_n^2 = s_1^2 - 2s_2$$

and that

$$\sum_{sym} u_1^2u_2 = s_1s_2 - ns_3$$

where the sum is the orbit sum of $u_1^2u_2$. Then,

$$s_1(u_1^2 + \cdots + u_n^2) = u_1^3 + \cdots + u_n^3 + \sum_{sym} u_1^2u_2$$

which simplifies as

$$u_1^3 + \cdots + u_n^3 = s_1(s_1^2 - 2s_2) - s_1s_2 + ns_3 = \boxed{s_1^3 - 3s_1s_2 + ns_3}$$

□

Problem. 3.1. Let f be a polynomial of degree n with coefficients in F and let K be a splitting field for f over F. Prove that $[K : F]$ divides $n!$.

Solution. First, suppose f is irreducible. Then we prove the claim by induction; the result for $n = 0$ is trivial ($K = F$). In general, if α is a root of f, then

$$[K : F] = [K : K(\alpha)][K(\alpha) : F] = n[K : K(\alpha)].$$

In $K(\alpha)$, f factors as $(x - \alpha)g(x)$; if f_1 is an irreducible factor of g (in $K(\alpha)$), then, letting L_1 be the splitting field of f_1, we have by hypothesis that $[L_1 : K(\alpha)]$ divides $(\deg f_1)!$. Then we can take an irreducible factor f_2 of g/f_1 and take the splitting field L_2/L_1, and so on, giving a chain of extensions $K = L_k \supseteq \cdots \supseteq L_1 \supseteq K(\alpha) \supseteq K$. The degrees of the f_is add up to $n - 1$, so that

$$[K : K(\alpha)] = [L_k : L_{k-1}][L_{k-1} : L_{k-2}] \cdots [L_2 : L_1][L_1 : K(\alpha)] = n_k!n_{k-1}! \cdots n_2!n_1! \mid (n-1)!$$

where the divisibility is a result of the multinomial coefficient $\binom{n-1}{n_1,\ldots,n_k}$ being an integer; then the factor of n from $[K : K(\alpha)]$ gives divisibility by $n!$.

For possibly reducible f, let g be an irreducible factor of f in F. Then the splitting field K_g of g has degree dividing $(\deg g)!$ by the irreducible case and $[K : K_g]$ divides $(\deg f/g)! = (\deg f - \deg g)!$ by induction, so

$$[K : F] \mid (\deg g)!(n - \deg g)! \mid n!$$

with divisibility coming from $\binom{n}{\deg g}$ being an integer. □

Problem. 6.1. Let α be a complex root of the polynomial $x^3 + x + 1$ over $\mathbb{Q}$, and let K be a splitting field of this polynomial over $\mathbb{Q}$. Is $\sqrt{-31}$ in the field $\mathbb{Q}(\alpha)$? Is it in K?

Solution. $\sqrt{-31}$ is not in $\mathbb{Q}(\alpha)$: if it were, then we would have

$$3 = [\mathbb{Q}(\alpha) : \mathbb{Q}] = [\mathbb{Q}(\alpha) : \mathbb{Q}(\sqrt{-31})][\mathbb{Q}(\sqrt{-31}) : \mathbb{Q}] = 2[\mathbb{Q}(\alpha) : \mathbb{Q}(\sqrt{-31})],$$

which is impossible because the degree of the extension is an integer.

However, $\sqrt{-31}$ is in K, because $\sqrt{-31}$ is the square root of the discriminant $-4(1) - 27(1) = -31$ of $x^3 + x + 1$. □

Problem. 7.2. Let K/F be a Galois extension such that $G(K/F) \cong C_2 \times C_{12}$. How many intermediate fields L are there with (a) $[L : F] = 4$, (b) $[L : F] = 9$, (c) $G(K/L) \cong C_4$?

Solution. Because K/F is Galois, $24 = |G(K/F)| = [K : F]$.

(a) $[L : F] = 4$ if and only if $[K : L] = 6$. Such fields L correspond bijectively to subgroups of $C_2 \times C_{12}$ with order 6. Noting that all the subgroups are abelian (because the group itself is abelian), every order 6 subgroup of $G(K/F)$ must be isomorphic to C_6. The order 6 elements in $C_2 \times C_{12}$ are

$$(1, 2), (1, 4), (1, 8), (1, 10), (0, 2), (0, 10),$$

of which $(1, 2)$ and $(1, 10)$ generate the same subgroup, $(1, 4)$ and $(1, 8)$ generate the same subgroup, and $(0, 2)$ and $(0, 10)$ generate the same subgroup. Therefore $C_2 \times C_{12}$ has three subgroups of order 6, so there are $\boxed{3}$ fields L with the desired property.

(b) $[L : F] = 9$ is impossible, since this would imply $[K : L] = 24/9 \notin \mathbb{Z}$.

(c) Each subgroup isomorphic to C_4 corresponds to exactly one intermediate field with Galois group isomorphic to C_4 (with the field being the fixed field of the subgroup, as per the Galois correspondence). We therefore need to count the number of subgroups isomorphic to C_4. These are generated by elements of order 4, of which there are four:

$$(0, 3), (0, 9), (1, 3), (1, 9).$$

The first two generate the same subgroup and the latter two generate the same subgroup, giving $\boxed{2}$ subgroups of $G(K/F)$ isomorphic to C_4. □

Problem. 10.3. Let $\zeta = \zeta_7$. Determine the degree of the following elements over $\mathbb{Q}$:

(a) $\zeta + \zeta^5$,

(b) $\zeta^3 + \zeta^4$,

(c) $\zeta^3 + \zeta^5 + \zeta^6$

Solution. (a) The degree is $\boxed{6}$.

We know that the subfield generated by $\zeta + \zeta^5$ is a subfield of $\mathbb{Q}(\zeta)$, and that the Galois group of $\mathbb{Q}(\zeta)$ is the set of automorphisms $\zeta \mapsto \zeta^k$ for $k = 1, \dots, 6$, which we will denote with ϕ_k. This means that $\zeta + \zeta^5$ shares its minimal polynomial with

$$\begin{aligned} \phi_1(\zeta + \zeta^5) &= \zeta + \zeta^5 \\ \phi_2(\zeta + \zeta^5) &= \zeta^2 + \zeta^3 \\ \phi_3(\zeta + \zeta^5) &= \zeta^3 + \zeta^1 \\ \phi_4(\zeta + \zeta^5) &= \zeta^4 + \zeta^6 \\ \phi_5(\zeta + \zeta^5) &= \zeta^5 + \zeta^4 \\ \phi_6(\zeta + \zeta^5) &= \zeta^6 + \zeta^2, \end{aligned}$$

and no other complex numbers (because $\mathbb{Q}(\zeta)$ is a Galois extension, hence a splitting field). Recalling that $\zeta, \dots, \zeta^6$ are a basis for $\mathbb{Q}(\zeta)$, we can conclude that $\phi_i(\zeta + \zeta^5) = \phi_j(\zeta + \zeta^5)$ cannot hold for $i \neq j$, since this would imply linear dependence between at most four powers of ζ. This means that the minimal polynomial of $\zeta + \zeta^5$ has six distinct roots, i.e. the degree of $\zeta + \zeta^5$ is 6.

(b) The degree is $\boxed{3}$. Letting $x = \zeta^3 + \zeta^4$, we can see that (noting $\zeta^3 \cdot \zeta^4 = 1$)

$$x^2 - 2 = \zeta + \zeta^6, \quad x^3 - 3x = \zeta^2 + \zeta^5.$$

Therefore

$$x^3 + x^2 - 2x - 1 = 1 + x + x^2 - 2 + x^3 - 3x = 1 + \zeta + \zeta^2 + \cdots + \zeta^6 = 0$$

and the polynomial on the LHS doesn't factor because it has no rational roots (as seen by applying the rational root theorem and checking the only two candidates ± 1). We have shown that x is the root of an irreducible polynomial with degree 3, which is what we wanted.

(c) The degree is $\boxed{2}$. If $x = \zeta^3 + \zeta^5 + \zeta^6$, then

$$x^2 = \zeta^6 + \zeta^3 + \zeta^5 + 2(\zeta + \zeta^2 + \zeta^4) = -2 - x.$$

Therefore the minimal polynomial of x is $x^2 + x + 2$, so x has degree 2 over $\mathbb{Q}$. ∎